1.Supervised Learning: Supervised learning is a method where the model is trained using labeled data, meaning both input (X) and correct output (Y) are provided. The model learns the relationship between input features and target variable.

2. Unsupervised Learning: Unsupervised learning is used when data has no labeled output. The model tries to discover patterns or structures in the data.

3. Semi-Supervised Learning: Semi-supervised learning uses a small amount of labeled data and a large amount of unlabeled data.

4. Reinforcement Learning: Reinforcement learning is a learning method where an agent learns by interacting with an environment and receiving rewards or penalties. The goal is to maximize total reward over time.

Local Minimum: A local minimum is a point where the function value is lower than the nearby points around it, but it may not be the lowest value in the entire function.

Local Maximum: A local maximum is a point where the function value is higher than nearby points, but it may not be the highest value overall.

Global Minimum: The absolute lowest point in the entire function.

- When training ML models, we minimize a loss function (error).
- Optimization algorithms like Gradient Descent try to find the minimum loss.

Gradient Descent:

Gradient Descent is an optimization algorithm used in machine learning to minimize the loss (error) of a model by adjusting its parameters step-by-step.

$y = mx + c$

The error is measured using a **loss function**.

Loss Function:

A common loss function in regression is **Mean Squared Error.**

Learning Rate:

Learning rate determines **how big the steps are**.

Bias and variance:
Bias
Bias is the error caused by oversimplifying the model.
It happens when the model makes strong assumptions about the data and cannot capture the real pattern.

- Model is **too simple**

- Cannot learn complex relationships
- Leads to **underfitting**

Variance
Variance is the error caused by the model being **too complex**.
The model learns the **training data too well**, including noise.
- Model is **very complex**
- Fits training data perfectly
- Performs poorly on new data

This is called **overfitting**.

Bias–Variance Tradeoff:
The goal is to **balance bias and variance**.
If we decrease bias too much, variance increases.
If we decrease variance too much, bias increases.

Multicollinearity:
Multicollinearity occurs when two or more independent variables in a regression model are highly correlated with each other.
This means the variables contain similar information, which makes it difficult for the model to determine which variable is actually influencing the target variable.

Detecting Multicollinearity using Pearson Correlation
One common method is Pearson Correlation Coefficient.
It measures the linear relationship between two variables.

Correlation Heatmap
A heatmap visually shows correlation between variables.

Heatmap colors represent correlation strength.
Red - High positive correlation
Blue - Negative correlation
White - Low correlation

If If correlation between two independent variables is $|r| > 0.8$
- remove one feature
- combine features

Remedies for Multicollinearity
1. Do Nothing (When Multicollinearity is Weak)

2. Remove One of the Correlated Variables
If two variables carry almost the same information, remove one.
3. Combine Correlated Variables
Sometimes instead of removing variables, we combine them into a single feature.
4. Other Common Methods (Often Mentioned)
Principal Component Analysis (PCA)
Transforms correlated variables into **uncorrelated components**.
Regularization (Ridge Regression)
Adds a penalty term to stabilize coefficients.

Regularization (in Linear Regression / ML)
Regularization is a technique used to reduce overfitting by adding a penalty to large model weights (coefficients).
When a model becomes too complex, it may memorize training data. Regularization forces the model to keep coefficients smaller, making the model simpler and more generalizable.
Lasso Regularization (L1 Regularization)
Lasso adds a penalty equal to the absolute value of coefficients.
Lasso penalizes large weights and can shrink some coefficients completely to zero.
This means it automatically performs feature selection.
Ridge Regularization (L2 Regularization)
Ridge adds a penalty using squared coefficients.
Ridge reduces coefficients but does NOT make them exactly zero.
Regularization helps to:

- Reduce **overfitting**
- Handle **multicollinearity**
- Improve **model generalization**
- Perform **feature selection** (Lasso)

Polynomial Regression

- Polynomial Regression is a type of regression used when the relationship between the independent variable (X) and dependent variable (Y) is not linear (not a straight line) but curved.
- The **degree** determines how flexible the curve is.